Agriculture: Apple Pest Management Guidelines

European Canker

Nectria galligena

Symptoms and Signs

In fall or spring European canker causes reddish brown lesions(/PMG/N/D-AP-NGAL-BT.004.html) to appear on small branches just below leaf scars. These elongate into cankers with concentric ridges and may cause dieback of shoots in spring. Calyx rot of fruits can occur in years when rain precedes harvest. Pruning wound infections are seen occasionally (especially on the Delicious cultivar). On superficial examination, such infections may be confused with fire blight.

Comments on the Disease

This fungus survives in old bark cankers and produces spores that enter fresh leaf scars during fall rains. The Delicious variety is most susceptible, followed by Gravenstein and Rome Beauty. This disease is worse in the Sebastopol area of Sonoma County in years with prolonged fall rains.

Management

European canker is managed primarily by pruning and protectant fungicides. Cankers should be pruned out of trees as they ultimately kill branches and also serve as sources of inoculum. Prune and burn diseased wood early in summer. At this time symptoms are obvious and spread of the fungus is not likely. Because infection occurs through leaf scars and leaves fall over a long period, two treatments are necessary each fall to protect new leaf scars.

Organically Acceptable Methods

Treatments with Bordeaux or approved fixed copper materials are organically acceptable.

Treatment Decisions

If European canker is damaging your orchard, apply a freshly prepared Bordeaux mixture of 10:10:100 or a fixed copper material at label rates during early leaf fall, before rains begin. Where the disease is serious, make a second application when three-fourths of the leaves have fallen.

	Common name	Amount to use	REI‡	PHI‡
	(Example trade name)		(hours)	(days)

Pesticide precautions

Protect water

Calculate VOCs

Protect bees

Not all registered pesticides are listed. The following are ranked with the pesticides having the greatest IPM value listed first—the [most effective(/agriculture/apple/Fungicide-Efficacy-for-Apple-Diseases/)] and least likely to cause resistance are at the top of the table. When choosing a pesticide, consider information relating to the pesticide's [properties(/agriculture/apple/General-Properties-of-Fungicides-Used-in-Apples/)] and [application timing(/agriculture/apple/Treatment-Timings-for-Key-Diseases/)], [honey bees(/bee-precaution-pesticide-ratings/)], and [environmental impact(/mitigation/)]. Always read the label of the product being used.

A.	**BORDEAUX MIXTURE#**			
	10:10:100	Label rates	See comments	See comments
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Multi-site contact (M1)			
	COMMENTS: When used on organically grown produce, all ingredients must be certified by the organic grower's certifying agent. Observe the most restrictive label precautions and limitations among all the Bordeaux mixture ingredients, including the most restrictive PHI and REI. For more information on creating a Bordeaux mixture, see *UC IPM Pest Note: Bordeaux Mixture*(/PMG/PESTNOTES/pn7481.html), ANR Publication 7481.			
B.	**FIXED COPPER#**	Label rates	24	0
	MODE-OF-ACTION GROUP NAME (NUMBER[1]): Multi-site contact (M1)			
	COMMENTS: Not all copper compounds are approved for use in organic production; be sure to check individual products.			

#	Acceptable for use on organically grown produce.
1	Group numbers are assigned by the [Fungicide Resistance Action Committee (FRAC)(http://www.frac.info/)] according to different modes of actions. Fungicides with a different group number are suitable to alternate in a resistance management program. In California, make no more than one application of fungicides with mode-of-action Group numbers 1,4,9,11, or 17 before rotating to a fungicide with a different mode-of-action Group number; for fungicides with other Group numbers, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action Group number.

UC PEER REVIEWED

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